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From Data to Podium: A Machine Learning Model for Predicting Formula 1 Compound Decisions

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Abstract

This thesis explores the optimization of Formula 1 pit stop strategies, integrating advanced analytics and machine learning to predict tire compound decisions. A novel aspect of this study is the clustering of driver profiles based on performance, tactical, and behavioral metrics, which provides a deeper understanding of driver characteristics and their impact on race strategy. By analyzing data from the FastF1 API and employing various machine learning techniques, we developed predictive models that not only forecast compound decisions with higher accuracy but also highlight the significance of personalized strategies tailored to different driver clusters. The findings demonstrate the potential of combining driver clustering with predictive analytics to refine pit stop strategies, offering teams a competitive edge through data-driven decision-making.

Keywords

Data, Data Analytics, Sports Analytics, Formula 1, Strategy, Prediction Model, Machine Learning

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1. Introduction

In the competitive world of sports, the strategic use of analytics has revolutionised how competition is understood and won. This transformation is particularly evident in Formula 1, a sport where the smallest decisions can have significant impacts; leveraging data effectively is not merely an advantage but a cornerstone of success. Central to these strategies are the critical decisions made during pit stops, which can significantly influence race outcomes. In this context, the role of advanced analytics has become increasingly prominent, offering new avenues to optimize these crucial decisions.

While research has been conducted on various aspects of Formula 1 strategy, a notable gap exists in exploring driver clustering and its potential impact on strategic decisions as well as the evaluation of analytical approaches. This thesis is motivated by the prospect of harnessing advanced analytics and driver clustering to add additional insights to pit-stop strategies. The aim is to explore how a refined understanding of driver behaviour can enhance race outcomes and the interpretation of analytical outcomes when integrated with data-driven approaches.

This study is guided by the research question: "*How can the application of advanced analytics and driver clustering in Formula 1 enhance the accuracy and effectiveness of strategic decision-making, particularly in the context of optimising pit stop strategies?*" The objectives are twofold: firstly, to develop a robust clustering model based on performance, tactical, and behavioural metrics; and secondly, to use the results of this model within analytical frameworks, like prediction models, to assess its impact on compound selection decisions.

Our methodology centres around leveraging data mainly from the FastF1 API, focusing on recent seasons to construct a clustering model, providing a multifaceted perspective on driver profiles. The methodology extends to applying this clustering within different analytical models, aiming to shed light on the intricacies of pit-stop strategies and potentially add optimisations or additional insights.

This research aims to contribute significantly to the fields of sports analytics and Formula 1 strategic planning. By offering a novel perspective on driver behaviour and its implications for race strategy, the findings of this study could potentially guide teams and drivers towards more informed and effective decision-making. The insights gained could not only enrich academic discourse but also provide practical tools for strategic optimisation in the high-pressure environment of Formula 1 racing.

To lay the foundation for our analysis, we begin with a comprehensive literature review that covers the broader spectrum of sports analytics, with a specific focus on data analytics in Formula 1. This review will also discuss the application of clustering in other sports, noting its relatively nascent presence in Formula 1 literature. Following this, we detail our methodology, including our full data collection process and the clustering of Formula 1 drivers based on their performance metrics. In subsequent chapters, we examine tire compound decisions made during races and explore how these elements interact with the identified driver behaviour clusters.

2. Literature Review

2.1. Evolution and Scope of Sports Analytics

Sports analytics refers to the application of scientific techniques for investigating and modelling sports performance. It entails organising historical data in a structured manner, applying predictive analytical models to the data, and utilising information systems to inform decision-makers (Morgulev, Azar, and Lidor 2018). This practice enables sports organisations to secure a competitive advantage through enhanced player performance analysis, game strategy formulation, health and injury prevention, fan engagement, and operational strategy optimisation (Nadikattu

2020; Tan 2023). Originating in the 1960s with notational analysis in sports like American football and basketball (Hughes and MFranks 2004), sports analytics has evolved significantly with technological advancements and the rise of big data. Today, its application extends across various sports domains, from individual sports like tennis to team sports such as football and has become central in the data-driven world of motorsports like Formula 1. This wide adoption underscores the broad reach and applicability of sports analytics across diverse sporting disciplines (Bai and Bai 2021).

As technology evolved, sports analytics underwent a significant transformation, unveiling new avenues for analysing and improving sports performance. A recent comprehensive review by Ghosh et al. (2023) classifies the technological advancements in sports analytics into three primary research fields: sensors, computer vision, and wireless and mobile-based applications. These areas form the bedrock of modern sports analytics, providing essential methods to collect, analyse, and interpret data. These technological advancements prove particularly pertinent in Formula 1, a sport renowned for its data-driven approach. Incorporating hundreds of sensors in racing cars facilitates real-time data collection, covering various parameters such as speed, temperature, and throttle percentage (Shapiro 2023). Effective analysis of this data offers invaluable insights for making informed decisions on pit stop strategies, car setups, and race strategies. The inclusion of artificial intelligence (AI) and machine learning (ML) algorithms amplifies the potential of these technological advancements, enabling more sophisticated analysis and predictive modelling (Ghosh et al. 2023; Dindorf et al. 2023). Integrating these novel technologies with sports analytics methodologies has opened and revolutionised research opportunities in Formula 1 racing.

2.2. Analytical Approaches in Formula 1 Racing

As established in the preceding section, sports analytics is central to enhancing competitive strategies across various sports disciplines. This becomes particularly evident in Formula 1, a sport where even the minutest decision can significantly alter race outcomes. In F1, where the stakes are high and the financial implications are vast, the synergy between data analytics and elite sporting performance exemplifies the comprehensive capabilities of sports analytics. The body of literature directly engaging with sports analytics in the context of Formula 1 is limited both in terms of the quantity of research and the range of thematic focus. Broadening the search parameters to include 'Circuit Racing Motorsports'—thereby encompassing NASCAR, Formula E, and similar series—yields an expanded body of work. Preliminary examination suggests that while there has been increasing interest in this field, it appears that the field has not yet reached a point of saturation, indicating sufficient opportunity for further research and contribution.

The existing literature can be divided into several overarching categories, however, two are predominant: lap time simulations and race simulations. As Heilmeier (2018) highlights, it is crucial to differentiate between race simulations and the more prevalent lap time simulations. The latter predominates in the literature and typically focuses on the physical or engineering aspects rather than on the holistic view of an entire race. Siegler (2000) identifies three distinct approaches to lap time simulations: Steady State, Quasi-Static, and Transient. Heilmeier (2019) published a study on quasi-static lap time simulation, applying it to both Formula 1 and Formula E to illustrate its utility. Colunga (2014) examined the modelling of transient cornering and suspension dynamics, along with the investigation of control strategies for an ideal driver within a lap time simulation framework. In a similar vein, Timings (2014) contributed to this body of work by aiming to develop a robust lap time simulation, referring to its comprehensive nature and its resilience in varying conditions.

However, for this thesis, race simulations and their components, specifically those that prioritize pit stop strategies as a key component in modelling or predicting race outcomes, are of greater relevance. Such simulations are instrumental in forecasting final standings by accounting for various factors, including driver interactions, empirical fuel consumption models, tire wear, and probabilistic effects. Bekker (2009) developed one of the earlier holistic race simulations to replicate key on-track activities in Formula 1, such as mechanical failures, overtaking manoeuvres, and pit stops. This model facilitates strategy planning by simulating the mechanical and physical dynamics of a race, thereby offering a team a potential advantage. More recently, Heilmeier (2018) outlined a simulation methodology for circuit motorsport racing strategies, which considers variables such as pit stops, tire choices, and tire degradation. The tool is designed to rapidly simulate races based on discrete lap data and adjustable strategy inputs. Building on this previous work, Heilmeier (2020) introduces a new further improving the simulation. This advanced version surpasses simple optimisation models by providing a comprehensive, automated simulation that responds in real-time to race dynamics. Heilmeier (2020) suggests that the current methodology for optimising pit stop decisions and associated tire compound selections might benefit from additional exploration in the future. Furthermore, he acknowledges the omission of complex strategies, such as the undercut—a tactic where a driver pits and switches to faster tires to gain time on rivals who pit later—from current models. He advocates for the integration of such tactics into subsequent models to enrich the decision-making process.

In addition to comprehensive racing simulations, focused research activities are directed at optimising specific components of the racing domain and simulations themselves, such as pit stops. These efforts aim to refine these aspects to their utmost efficiency. One research exemplifies this by employing machine learning algorithms to aid tire strategy decisions in the NASCAR series.

This work utilises predictive analytics, employing historical race data to forecast positional shifts consequent to variables such as tire change frequency and tire lifespan. Extensive feature testing has revealed that support vector regression and LASSO regression yield the highest accuracy in results (Tulabandhula and Rudin 2014). Additionally, Bell (2016) advances the analysis by conducting a comprehensive analysis of performance determinants in Formula 1, examining the evolving contributions of team dynamics and driver skills over time. The study results in a systematic ranking of drivers, offering insights into the qualifications of the 'potential best' based on a quantifiable set of criteria. Furthermore, Monte Carlo methods and analysis of probabilistic factors play a significant role in simulating the inherent variability present in lap times, pit stops, race incidents, and potential degradation of vehicle parts. The study of Heilmeier (2020) builds upon this foundation, providing a comparative analysis of the seminal works of Bekker (2009), Phillips (2014), and Salminen (2019), thereby extending the understanding of these stochastic elements in race simulations.

The existent body of research on Formula 1 is mainly based on the scope of publicly available data, which has been limited to lap times, and race outcomes. Such data has predominantly been sourced from the Ergast API, a privately maintained database for Formula 1 statistics (Ergast 2009). However, the introduction of the Fast F1 API marks a significant progression in data availability, offering not only the information provided by the Ergast API but also a more comprehensive set of F1 data. This includes telemetry data, official weather statistics, and track information (FastF1 2020). The introduction of the Fast F1 API thus promises to broaden the scope of current literature and models by facilitating the incorporation of mechanical details of the vehicle (such as current gear, RPM, and speed) and more granular data like positional coordinates, distances between drivers, and time specific air and track temperatures. The newly available data points, particularly telemetry data, which have received little attention in the scientific literature to date, present potential opportunities to enhance and broaden current analytical frameworks. The richer and more granular nature of this data allows for a more detailed examination of specific drivers' behaviours based on positional and telemetry information. A prospective methodological approach might include clustering drivers according to their behavioural patterns in various racing scenarios. Such clustering could yield valuable insights into performance differentiators and decision-making processes throughout a race and its strategic decisions.

2.3. Clustering Techniques in Sports Analytics and Formula 1

Clustering, a core technique in unsupervised machine learning, groups data points into distinct categories based on common attributes without predefined labels (Pedregosa et al. 2011). In sports analytics, clustering is a key tool, enabling teams and coaches to decode complex patterns and detect subtle correlations. This method can help identify performance trends and effective team compositions, which, in turn, can be leveraged to improve predictive models that forecast future sports outcomes based on the grouping of player or play characteristics. Clustering may uncover non-intuitive groupings or strategies, providing a strategic advantage in the competitive world of professional sports. Its utility and adaptability across various sports disciplines are well-documented, with a significant body of literature.

In Basketball analytics, player assessment and categorisation have evolved well beyond the confines of traditional position labels. One study by Duman, Sennaroğlu, and Tuzkaya (2021) applies hierarchical cluster analysis to a rich dataset of game-related statistics from 15 NBA seasons, uncovering four to six distinct playing styles within each traditional position. The clusters, characterised by unique attributes and skill sets, offer a multifaceted perspective on player capabilities, yielding strategic insights for player placement and team composition. Muniz and Flamand (2022) introduce an advanced clustering technique based on weighted networks. The methodology starts with k-means clustering to form preliminary groupings, which then inform a network where players are interconnected by weighted edges reflecting performance similarities. Employing the Louvain method for community detection, the study identifies eight player archetypes, surpassing the insight offered by the five traditional positions. They further enhance this method by using tracking data, adding a layer of depth to the analysis with precise measurements of player movements and interactions. For Football, a fuzzy clustering model that can handle mixed data types has been introduced. This model assigns objective weights to attributes such as player performance metrics, positional data, and physical characteristics, thereby uncovering clusters that the complexity of mixed-attribute data might hide. Such detailed analysis facilitates the identification of player clusters according to their on-field roles, skill sets, and physical profiles, which is instrumental in tactical team structuring and player market valuation (D'Urso, De Giovanni, and Vitale 2023). A study in tennis clustered 1188 Grand Slam players by analysing their physical and play style data, revealing four distinct profiles. Through two-step cluster analysis and further MANOVA and discriminant analysis, the research identified how factors like height and handedness correlated with performance, notably in serving and net play (Cui et al. 2019). Clustering in sports analytics reaches beyond mainstream sports, extending its application to disciplines like badminton. Sinadia and Murwantara (2022) apply k-means and hierarchical agglomerative clustering to analyse badminton athletes' performances, identifying clusters based on game results and consecutive scoring. K-means clustering discerns four distinct performance clusters, with one particularly strong cluster associated with high scores and consecutive points, supported by similar results from hierarchical clustering.

Collectively, these studies showcase the breadth and depth that clustering techniques bring to sports analytics. They enable a sophisticated segmentation of athletes, offering a granular understanding that can guide coaching decisions, athlete development, and competitive strategy formulation. In the data-rich context of Formula 1, the application of clustering techniques to group drivers by various data-driven similarities could potentially yield insights that extend beyond traditional performance metrics. While such methods have provided detailed understandings in other sports disciplines, the extent to which they have been applied to driver analysis in Formula 1 remains limited. Most existing studies on driver performance within this sport have focused on race results without a significant exploration of specific clustering techniques that have been beneficial in broader sports analytics.

For instance, one pioneering study by Phillips (2014) presents a statistical model that quantifies the contributions of drivers and teams to performance, using championship points as the metric. In another notable work by Rockerbie and Easton (2021), they employ an econometric method to dissect the relative impact of driver skill versus car technology on race results, discussing the "80-20 rule" but primarily concentrating on race positions and outcomes without delving into driver-specific skill and performance metrics. Similarly, a recent study utilises a Bayesian multilevel rank-ordered logit model to analyse historical ranking data, aiming to distinguish between driver skill and constructor efficiency (Van Kesteren and Bergkamp 2023). However, like its predecessors, it does not extensively investigate drivers' characteristics or behaviour.

While providing valuable insights, the discussed studies often centre around broader performance metrics and race outcomes without specifically exploring the clustering of drivers based on their attributes or behaviours. This gap indicates an opportunity for further research into the application of driver clustering techniques within Formula 1. Exploring driver clustering, especially by using detailed telemetry data, expands the analytical horizon and could establish a foundation for in-depth examination of strategic components, such as pit stop tactics and other decision-making processes in the sport. Furthermore, the comprehensive use of telemetry data remains underutilised in existing models. Incorporating telemetry data into clustering analyses could reveal more profound insights into driver behaviour, driving styles, and performance metrics, which go beyond the conventional race outcomes and rankings.

3. Data

Given the analyzed theoretical background found in our extensive literature, we glided on to the next part of the thesis - the development of a clustering technique to identify different driver profiles in the field of Formula 1. Essential for our success is the data we rely on, which is why we performed an extensive data collection process, which will be presented in the next sections of this thesis.

3.1. F1DataFetcher

The already mentioned, publicly accessible API from FastF1 and the associated Python library can be used to generate the required data. In contrast to the previously common ERGAST API, the FastF1 API also includes interesting aspects such as weather, position and telemetry information in addition to the usual Formula 1 data. Telemetry information includes speed, revolutions per minute and much more. Position data not only informs about the exact position of a car on the racetrack but also provides information about direct duels by showing which driver is driving in front of a driver and how far away he is. This data can be a great asset and a clear advantage over existing literature. That's why the decision on FastF1 as the main source of data was made quickly. Using this API is relatively simple and works without any problems. Interested readers are referred to the detailed documentation provided by the operators (FastF1 3.1.6). However, the API has one limitation for efficient data collecting purposes: the user can only ever load one event of a session, be it qualifying, race, free practice, etc., from the API. This is not due to the user's authorisation but merely to the logic of the API. To get around this, a new logic was created.

The Python class we developed is called F1DataFetcher. This allows us to pull data for multiple races simultaneously from the API into a notebook, automatically creates the required data frames in the required structure and outputs them as pandas data frames. In addition to the challenge of being able to receive multiple data at once, it was also important to be able to save the data so that local access without the need of reloading data every time was granted. Therefore, we created a relational database in PostgreSQL. This allows to save the individual data frames that are collected by the API via the F1DataFetcher in a meaningful, coherent, and comprehensible way and to keep them ready for further use. The database was created manually in pgAdmin4, and a connection was established via the Python library 'sqlalchemy'. This package allows users to send SQL commands in Python to SQL-based databases and receive pandas data frames. The methods of the FastF1 Data Fetcher were enhanced by the capability of sending collected data automatically to a specified PostgreSQL database. The final relational database contains a total of 10 tables. The structure and relations are composed as shown in Figure 1.

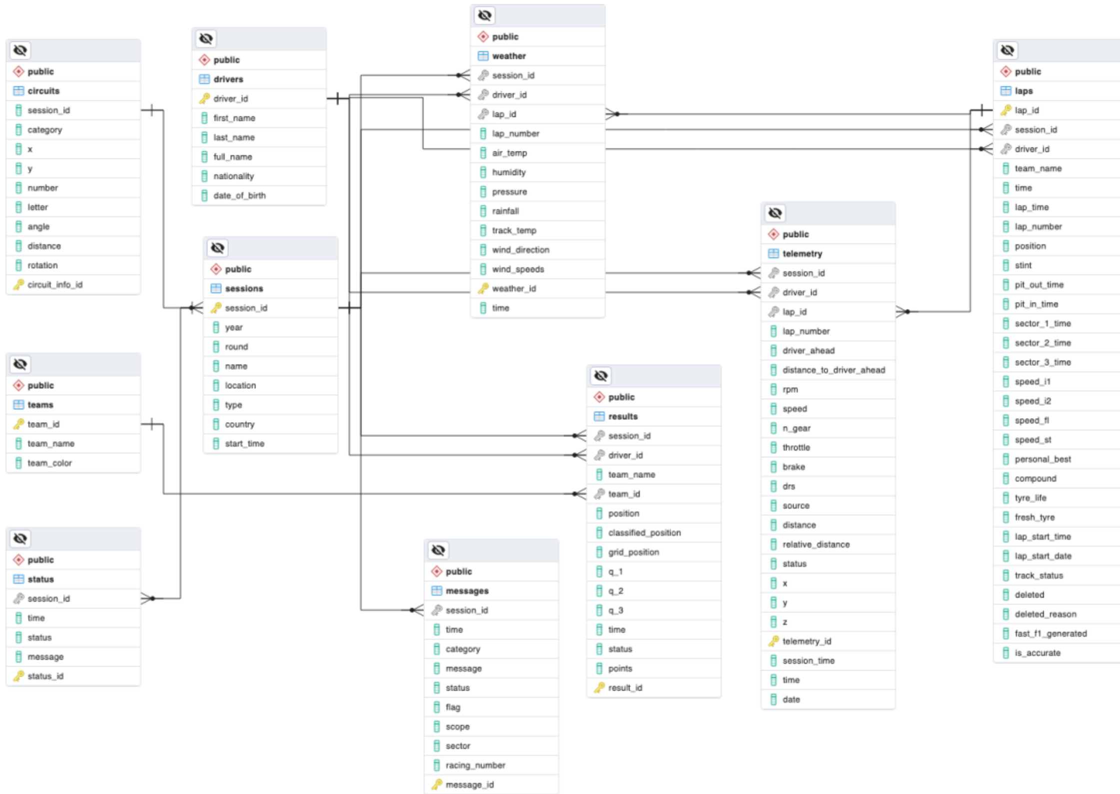

Figure 1: Simplified Schema of Our Own Data Base

The final F1DataFetcher class therefore contains ten methods plus one more to collect different data frames from FastF1 (and ERGAST API) and optionally store them in a database with the above schema. The data frames are on different levels which will be explained in the following along with the methods used to collect this data.

The first group of methods collect general information, such as drivers or teams. Although this data is collected on session level, the information is overarching and not tied to individual sessions. A comprehensive list of the methods used to collect this data is shown in Table 1.

Table 1: F1DataFetcher Methods - Group 1

Method	Description
`get_multiple_sessions()`	Load detailed information on grand prix events, optionally store to database
`get_unique_drivers()`	Load detailed information on unique drivers, optionally store to database
`get_unique_teams()`	Load detailed information on unique teams, optionally store to database

The second group of data is the one where data is on session level. The methods that collect these kinds of data are listed in Table 2.

Table 2: F1DataFetcher Methods - Group 2

Method	Description
`get_race_results()`	Load detailed information on grand prix results, optionally store to database
`get_race_control_messages()`	Load detailed information on race control messages, optionally store to database
`get_track_status()`	Load detailed information on flags and statuses, optionally store to database
`get_circuit_info()`	Load detailed information on track characteristics, optionally store to database
`get_weather()`	Load detailed information on weather characteristics, optionally store to database

The final set of methods collect data that is highly granular, meaning collected individually on lap basis for each driver. A full overview is provided in Table 3.

Table 3 F1DataFetcher Methods - Group 3

Method	Description
`get_laps()`	Load detailed information on laps, optionally store to database
`get_telemetry()`	Load detailed information on telemetry, highly granular
`store_to_postgresql()`	Store highly granular telemetry data to database

The methods of the F1DataFetcher provide a complete, easy to use way of collecting large amounts of data from FastF1 API and automatically store in PostgreSQL. They were then applied to set up a full database that served as the input data for the complete work.

3.2. Limitations

For feature engineering, data pre-processing and normalization, data was limited for better performance and other reasons. The process of data filtering was conducted with the following steps:

- **Exclusion of races during rain**: Grand Prix events in rainy weather conditions were excluded. The reason behind this was that in the complete database, there is a lack of data for events in the rain. Since not all drivers participated in all the races, and metrics can be significantly different on wet tracks, the decision was made to eliminate these data points from the data that serves as input for feature engineering.
- **Removal from retired drivers**: Although there is some information in the data about why they retired, there is no sufficient way to know if it was their fault, possibly some other driver's fault, a tactical decision or simply a problem with the car. Therefore, the decision was made to remove all data of drivers who retired on a particular track and only look at those data points where a driver successfully drove all laps of the race.
- **Removal of interrupted laps**: Sometimes races get interrupted due to accidents, weather conditions or other reasons. In these cases, the data is noisy, leading to lap- and pit stop times of several hours, which makes no sense and only distorts some of our engineered features. To avoid these kinds of complications, the data for laps in which a red flag was hissed, meaning the race was interrupted, was omitted from the data.

4. Clustering

4.1. Introduction to Clustering in Racing Analysis

Transitioning to clustering, a principal method in unsupervised machine learning, we leverage the refined dataset to explore and interpret the complex dynamics of Formula 1 racing.

Clustering is essential in the analysis and interpretation of complex data sets. This technique, as elucidated by Jain (2010), involves the organization of a collection of patterns into clusters, based on similarity. Patterns within a cluster exhibit a higher degree of similarity to each other than to those in different clusters, thereby revealing the intrinsic structure of the data. The significance of clustering extends beyond mere data categorization. Jain (2010) emphasizes its role in exploratory data analysis, summarization, and compression, as well as its utility as a tool for hypothesis generation and testing. In diverse fields such as image and pattern recognition, bioinformatics, and information retrieval, clustering is instrumental in uncovering hidden structures, identifying anomalies, and simplifying complex data landscapes.

In the realm of sports analytics, particularly in Formula 1 racing the application of clustering takes on a more defined and critical role. In the high-octane world of Formula 1 racing, understanding the nuances of driver behaviour is essential for gaining a competitive edge. This study employs a clustering methodology to delve into the intricacies of driving behaviour, aiming to achieve a set of specific objectives that enhance our comprehension of what differentiates top performers on the track. A central goal of this research is the characterization of driver profiles through clustering. This involves categorizing drivers into distinct groups based on a comprehensive array of performance metrics and observed behaviours on the track. The aim is to construct detailed profiles that capture the essence of each driver's approach to racing, including their performance, behaviour, and tactics. These profiles are expected to provide a holistic view of each driver's strengths and areas for improvement, offering insights into the diverse strategies and techniques employed in

Formula 1 racing. Furthermore, the study engages in a comparative analysis across the derived clusters. This analysis is crucial for identifying the unique characteristics that distinguish each cluster, thereby shedding light on what sets apart the most successful drivers. This comparative approach is anticipated to reveal valuable insights into the factors that contribute to effective driving strategies and overall race performance.

Finally, the clustering analysis aims to provide practical strategic implications for teams and drivers. By understanding the distinct behavioural clusters, teams can tailor their training programs, strategy development, and in-race decision-making to better align with the identified strengths and weaknesses of their drivers. This objective is predicated on the belief that data-driven, personalized strategies can significantly enhance performance and competitiveness in Formula 1 racing. Aligning training and strategies with the insights gained from clustering analysis, teams can optimize their approach to each race, maximizing their potential for success in this dynamic and challenging sport.

4.2. Feature Engineering

Before being able to cluster driver behaviour and performance, it is vital to define metrics and input features that capture the essence of what is tried to analyse. Since Formula 1 is a complex competition with various influential factors, we decided to limit it down to three main aspects: The performance metrics employed to evaluate a driver's performance, behavioural metrics that illustrate drivers' conduct during driving, and tactical metrics, reflecting the strategic choices made by a driver or team throughout a race.

In the following section, an overview of the features created for the clustering of Formula 1 drivers is represented, detailing the data utilized for their calculation.

Performance Metrics

Despite the significant dependency of performance on the car's capabilities, for which we have insufficient data, our objective was to quantify specific dimensions of driver performance. We defined three main aspects of driver performance: lap time, position gain, and consistency in driving. The first two metrics are self-explanatory and do not require additional clarification. The third metric, consistency, aims to quantify a driver's uniformity on the track, specifically regarding lap time variations. These three metrics effectively represent a comprehensive assessment of a driver's performance.

Table 4: Clustering - Performance Metrics

Variable	**Name**	**Description**
Lap time	`avg_lap_time`	Average lap time for each driver on all races
Position gain	`position_gain`	Average number of positions gained (or lost)
Consistency	`consistency`	Standard deviation in average lap time

Behavioural Metrics

More complicated than defining metrics to see a driver's performance is mapping various aspects of driving behaviour to input features. The focus was primarily on aggressiveness and a driver's behaviour towards other drivers, especially the driver ahead of him. To analyse behaviour in direct competitive scenarios, we quantified various gaps and distances relative to the leading driver, thereby attempting to delineate patterns of conduct in these head-to-head duels. After all, the five new input features for capturing driver behaviour were defined as the *distance to the driver ahead*, the *time within range*, *persistence*, *overtakes*, and *defensive actions*.

Table 5: Clustering - Behavioural Metrics

Variable	Name	Description
Distance to driver ahead	`avg_distance`	Average distance to the opponent ahead
Time within range	`time_within_range`	Average time a driver spends in a range of 50 meters behind his opponent
Persistence	`persistence`	Standard deviation of the distance to the driver ahead
Overtakes	`overtakes`	Average number of successful overtakes
Defensive actions	`position_maintained`	Average number of laps without losing a position

Tactical Metrics

As tactical decisions that are made by team leads and the driver himself during constant communication play an essential role in Formula 1 and can impact a driver's final position and overall performance significantly, capturing differences in strategic decisions was the goal that was aimed for. Two main aspects of Formula 1 tactics are pit stop timing and compound choice. Deciding when to pit stop, which tires to wear and how many pit stops a driver does can drastically change the outcome of a race. To capture the strategic decisions for each driver, a new set of features was created from the data to capture the pit stop timing and the compound strategies.

Table 6: Clustering - Tactical Metrics

Variable	Name	Description
Pit time	`pit_time`	Average time a pit stop takes
Pit window: Early	`early`	Percentage of pit stops in first 25% of the laps
Pit window: Mid	`mid`	Percentage of pit stops in mid 50% of the laps
Pit window: Late	`late`	Percentage of pit stops in last 25% of the laps
Laps on SOFT	`laps_on_soft_percentage`	Percentage of laps driven on SOFT compound
Laps on MEDIUM	`laps_on_medium_percentage`	Percentage of laps driven on MEDIUM compound
Laps on HARD	`laps_on_hard_percentage`	Percentage of laps driven on HARD compound

4.3. Normalization

Normalization is a common technique to scale data to a standard range, eliminate redundant or noisy data**,** and improve algorithm performance significantly (Patel and Mehta 2011). Literature shows that it is common practice in machine learning to apply normalization techniques to data before using it as input data (Gopal, Patro, and Kumar Sahu 2015).

Since most clustering algorithms, such as k-means and others, rely on scaled data, we needed to normalize our data points. Another reason to follow that approach is that in motorsports in general and Formula 1, the information on the actual car build is highly confidential and due to this not publicly accessible. The cars have different characteristics not shared by the teams participating in Formula 1. There might be cars performing better in general than other cars. Since the focus was merely on the driving capabilities of the individual drivers only, a need to reduce the potential bias that could influence our clustering algorithms was urged.

Besides that, some input features can be significantly influenced by track characteristics, such as the number of corners on a track and their angle. Since data from various Grand Prixes was used, results achieved on multiple tracks were compared. To make them comparable, again, there is a need for data normalization.

There are several ways of scaling and normalizing data, the most common being z-score scaling and min-max normalization. For this purpose, the decision was made to apply the latter to our input data. Min-max normalization is a technique that keeps the original relationship among the data (Gopal, Patro, and Kumar Sahu 2015). It follows a simple calculation:

$$X_{scaled} = \frac{X - X_{min}}{X_{max} - X_{min}} \tag{1}$$

The Python library 'scikit-learn' (Pedregosa et al. 2011) provides a standard pre-processing algorithm for this kind of normalization that we applied to our data.

4.4. Selection of Clustering Algorithm

After normalizing our data to ensure comparability and reduce potential biases, as detailed previously, we move to the next crucial step in our analysis: the selection of an appropriate clustering algorithm. The choice of an algorithm is crucial, as it must align with the nature of our normalized data and the specific objectives of our study. The selection of the k-means clustering algorithm for the analysis of driver behaviour in Formula 1 racing is underpinned by several compelling reasons that align with the specific nature of the data involved. A primary factor is the simplicity and computational efficiency of k-means, as highlighted in the study of Jain (2010). This characteristic is particularly advantageous given the large volume and complexity of data generated in Formula 1. K-means offers an efficient and straightforward approach, which is essential for the initial exploratory analysis where a fundamental understanding of data grouping is sought. Additionally, the suitability of k-means for clustering numerical data, as noted by Arthur and Vassilvitskii (2007), aligns well with the mostly numerical nature of our engineered data. This alignment ensures that k-means is well-equipped to handle the specific types of data encountered in this research. In addition, the efficiency of k-means in processing large datasets is a significant advantage, as it allows for the swift and effective handling of the extensive data typical in Formula 1 racing. This efficiency, coupled with the ease of interpretation of its results, as Jain (2010) notes, makes k-means a practical choice for researchers and analysts who require clear and actionable insights from their data.

However, it must also be noted that k-means has certain limitations in its application. One challenge is that it operates under specific assumptions about the nature of the clusters it forms. Typically, the algorithm assumes that clusters are spherical and of similar size. However, as Arthur and Vassilvitskii (2007) point out, these assumptions may not always hold true in real-world data scenarios, including those encountered in Formula 1 racing. This limitation necessitates an understanding that the clusters formed may not perfectly represent the complex and varied nature of the data.

Additionally, k-means is known to be sensitive to the initial choice of centroids. This sensitivity can lead to variability in results across different runs of the algorithm, as observed by Celebi, Kingravi, and Vela (2013). This characteristic requires careful consideration and potentially multiple iterations to ensure robustness in the clustering outcomes. This need for precise selection of starting points leads directly into the next crucial step of our analysis: determining the optimal number of clusters.

4.5. Algorithm Parameters and Tuning - Determining the Number of Clusters

In defining the number of clusters for our k-means analysis, we employed a diverse approach to ensure methodological rigor. Initially, we utilized the elbow method, widely recognized for its effectiveness in cluster analysis (Kodinariya and Makwana, 2013). This method involved plotting the explained variance against the number of clusters and identifying an 'elbow' point where the rate of decrease in variance sharply changes. Our analysis indicated an elbow point between three and four clusters, suggesting that further increasing the number of clusters would result in marginal improvements in explained variance.

To complement this quantitative method, domain knowledge regarding the performance of drivers was also incorporated. This additional layer of analysis provided valuable context in guiding the decision-making process. Specifically, the decision to opt for four clusters was influenced by the distinct characteristics of drivers Russel and Latifi, who formed a separate fourth cluster, in contrast to the other three clusters. Given their different tire choices compared to the other drivers and their overall race positioning at the lower end of the grid, it was deemed appropriate to assign them to a separate cluster. This decision underscores the importance of integrating quantitative methods with domain-specific knowledge for more insightful and informed analyses in complex situations.

While the silhouette score, a measure of cluster cohesion and separation, was also calculated as part of our evaluation, it was the combination of all three together that ultimately led us to choose four clusters. This number of clusters appeared to offer a reasonable balance, ensuring that the clusters were distinct and meaningful without introducing overfitting or unnecessary complexity. The decision to opt for four clusters was made with the goal of achieving a segmentation that would allow for an insightful analysis of driver behaviour.

4.6. Results

Applying the outlined clustering methodology, drivers in the Formula 1 season of 2019, 2020, and 2021 were effectively segregated into four distinct clusters, which can be seen in the scatterplot in Figure 2. This is based on the presented combination of performance, tactical, and behavioural metrics.

The allocation of drivers across these clusters is as follows:

- **Cluster 0:** Albon, Raikkönen, Perez, and Magnussen.
- **Cluster 1:** Hamilton, Verstappen, and Bottas.
- **Cluster 2:** Grosjean, Kvyat, Leclerc, Norris, Ocon, Gasly, Ricciardo, Sainz, Stroll, Giovinazzi, and Vettel.
- **Cluster 3:** Latifi and Russell.

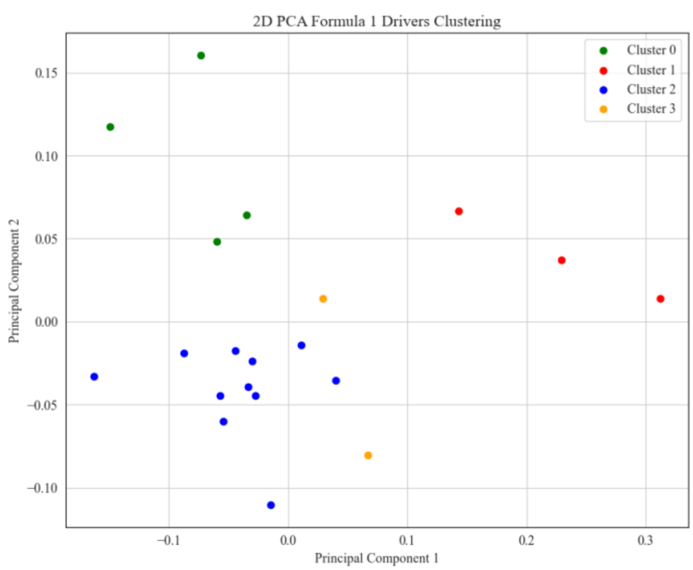

Figure 2: Scatterplot Illustrating Driver Clusters in 2D PCA Space

An initial analysis reveals a reasonable distribution reflective of known performance tiers within the sport. The top performers, typically dominating the front of the pack, are distinctly grouped in Cluster 1, while those frequently at the rear form Cluster 3. The midfield drivers, recognized through domain knowledge as a diverse group in terms of tactics and performance, are split between Clusters 0 and 2.

To describe the clusters in more detail and to interpret the classification, the underlying performance, tactical and behavioural metrics, which were defined as input features, need to be put into perspective. The different degrees of these features can also be seen in Figure 3. Here, the mean of the features per cluster are shown in the form of a parallel plot enabling a visual comparison between the clusters.

4.7. Interpretation

When interpreting the clustering results, it is important to recall that in Formula 1 the competition involves the top 20 drivers in the world. This elite group ensures a high degree of skill parity, as illustrated by our parallel plot analysis. The analysis underscores the close competition and high level of competence inherent in this sport. However, upon closer examination, subtle but meaningful variations in performance can still be observed. It's crucial to emphasize that while these differences are partly due to the varying budgets of the teams, and consequently the performance of the cars, the vehicle is not the sole determinant of a driver's overall performance. Teams with larger financial resources often have an advantage in terms of car performance, but the skills and decisions of the driver are also key to success. Teams with limited budgets may face challenges in vehicle performance, yet the individual performance and strategy of the driver can mitigate these disadvantages to some extent. This interplay between car performance and driver skill is a key element in analysing the competitive dynamics of Formula 1.

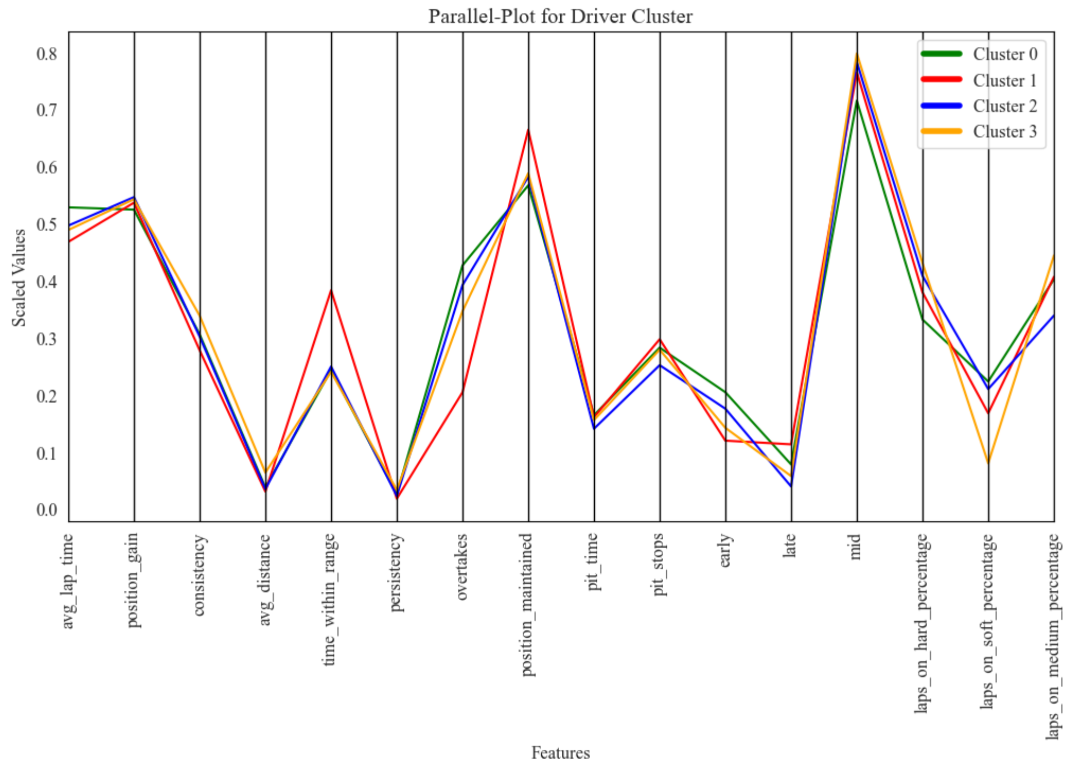

Figure 3: Parallel Plot Showing Mean Features of Driver Cluster

In our subsequent analysis, we will explore the specific characteristics of the clusters, bearing in mind the influence of vehicle differences, to provide a comprehensive understanding of the factors driving performance in Formula 1. This detailed examination starts with Cluster 0 and is based on the mean of each feature for each cluster.

Drivers in Cluster 0 are characterized by above-average lap times suggesting a focus on endurance and tactical positioning rather than outright speed. Their notable position gains reflect effective racing behaviour, with a high number of overtakes suggesting skill in dynamic racing scenarios. The moderate consistency of their performance can potentially be attributed to the positioning of the drivers in the midfield. Interactions between the drivers are much more frequent here, which has a greater impact on lap times. These interactions are also reflected in the frequent overtaking manoeuvres. The cluster's pit strategy, marked by average pit times and a preference for early stops, points to a tactical edge in race management. This is complemented by their significant usage of hard tires, implying a strategy centred on durability and long-term race planning over the immediate speed offered by soft tires.

In essence, drivers in Cluster 0 excel in strategic manoeuvring and consistency, prioritizing race longevity and tactical positioning over peak lap speed. This approach aligns with drivers who may not have the fastest cars but utilize strategic race management to optimize their overall standings.

Moving on to Cluster 1, which is characterised by the lowest average lap times, indicating superior performance. Their excellent position maintenance skills, combined with the highest average amount of pit stops, point to strategic control of both race pace and pit strategy. The frequent pit stops in this strategy are likely a tactical decision to optimize tire performance, considering there is usually sufficient time for an additional pit stop towards the end of the race without losing positions. This approach aligns with the balanced tire usage observed, where a high percentage of laps are run on both medium and hard compounds. Their exceptional consistency underscores a notable ability to deliver stable, reliable performances across various races. Additionally, their highest time within range of the driver in front, paired with a lower number of overtakes, suggests they often lead races and if not remain close to the driver in front, efficiently managing their positions and indicating an aggressive approach to overtaking manoeuvres. In essence, Cluster 1's drivers demonstrate a blend of speed, strategic expertise, and consistent performance, marking them as elite competitors in the Formula 1 field.

Next is Cluster 2, which presents a profile of competent race management with a focus on strategic tire usage. This cluster's average lap times and position gains, along with a consistency level akin to Cluster 0, indicate a balanced approach to race performance, matching speed with strategic positioning. The drivers in this cluster exhibit a lower frequency of overtakes compared to Cluster 0, yet they maintain their positions effectively during races. This suggests a focus on consistent lap performance and strategic placement rather than aggressive overtaking manoeuvres. Their ability to hold positions is indicative of skilful ability, particularly in defending against competitors and capitalizing on track opportunities. A notable aspect of Cluster 2 is their pit strategy. These drivers tend to pit the earliest among all clusters and have the lowest average number of pit stops. The fewer pit stops suggest a preference for longer stints on track. This aligns with their unique tire usage pattern, as Cluster 2 is the only group to perform more laps on hard tires compared to medium ones. This preference indicates a strategy leaning towards tire durability and longer race stints, possibly to maintain consistent lap times and reduce the time lost in pit stops. The hard tire's longevity would be particularly advantageous in races with high tire degradation or where preserving tire life is crucial for race strategy.

In summary, Cluster 2 drivers excel in strategic racing behaviour and tire management, combining average lap times with effective position holding and a distinctive early pitting strategy, highlighting a focus on endurance and tactical adaptability in the Formula 1.

Lastly, Cluster 3 presents a unique set of characteristics. Their lap times and position gains are comparable to those in Cluster 2, suggesting a similar approach in terms of speed and race advancement. However, the distinctive feature of this cluster is the lowest consistency among all groups, as indicated by the highest standard deviation in average lap times. This variability could be attributed to a range of factors such as adapting to different race conditions, varying strategic choices, or fluctuations in both driver and car performance. Another notable aspect of Cluster 3 is their racing style, which seems to focus on endurance and stability. This is inferred from their longest average distance to the driver ahead. Such characteristics may imply a tendency towards a more calculated and defensive racing approach, possibly prioritizing maintaining a steady pace and avoiding risks. The tire usage pattern in this cluster further supports this interpretation. The drivers in Cluster 3 have the lowest percentage of laps on soft tires, which are typically used for aggressive, short-term speed advantages. Their reluctance to use soft tires might indicate a preference for strategies that emphasize tire longevity and sustained performance over short bursts of speed. This approach is often seen in races where managing tire degradation and fuel consumption is critical for achieving a favourable outcome.

In conclusion, Cluster 3's drivers appear to adopt a strategic approach focused on endurance and stability, with less emphasis on aggressive manoeuvres and high-speed performance. Their lower consistency might reflect a more adaptive or variable strategy, responding to the specific demands of each race. This approach, combined with a cautious tire strategy, suggests a focus on long-term race management rather than immediate position gains.

In summary, the clustering analysis of Formula 1 drivers reveals distinct strategic profiles across the four groups. Cluster 0 and 2 both emphasize endurance and tactical positioning, but Cluster 2 distinguishes itself with an earlier pitting strategy and a greater emphasis on hard tires. Cluster 1 stands out for its combination of high speed, strategic pit stops, and consistent performance, showcasing the traits of front-running drivers. In contrast, Cluster 3 is defined by its variability in performance and a more conservative tire strategy, indicating a focus on stability and endurance. This comparative analysis highlights the diverse approaches in Formula 1, from aggressive speed and tactical prowess in Cluster 1 to the more calculated and endurance-focused strategies in Clusters 0, 2, and 3.

5. Optimizing Race Strategy: A Machine Learning Model for Predicting Formula 1 Compound Decisions

Building on the comprehensive review of literature and the development of driver profile clusters in the first part of our thesis, the second part takes a significant leap forward. Here, we harness the insights gained from clustering to craft a sophisticated prediction model focused on compound decisions in pit stop scenarios. This section not only demonstrates the application of our theoretical groundwork in a practical setting but also evaluates the effectiveness of these strategies across the varied driver clusters. By intertwining these two parts, our research offers a holistic view on optimizing Formula 1 performance, showcasing the synergy between driver characteristics and strategic decision-making in high-stakes environments.

5.1. Introduction

In Formula 1, selecting the right tire compound at the right time is crucial as it can impact a driver's final race position. Therefore, it plays a significant role in Formula 1 tactics. Per the rules, a driver must use at least two compound types during the race. Pirelli picks and supplies the available compounds for each Grand Prix. Since different compounds last a different number of laps, and tires wear out over time, determining the right time to apply which compound is a complex task in Formula 1 racing strategy. This article will present an approach to creating and improving automated data-based decision-making for the issue of compound choice during Formula 1 competitions. We trained several machine learning models to predict compound decisions in race scenarios, compared them, and implemented the best-performing one into an existing race simulation to test it in a real-world scenario.

5.2. Methodology

The procedure of developing predictive models that focus on compound decisions in pit stop scenarios is quite a demanding task that involves a lot of steps. A thorough explanation of every single step is provided in the methodology part of this chapter and is presented in detail in the following sections.

5.2.1. Data Acquisition

As a base for this work, we took data from a public repository containing an SQLite database file made accessible by Heilmeier (2020). The database was built using ERGAST API (Ergast 2009) to collect information for the Formula 1 season from 2014 to 2019 (Heilmeier, Thomaser, et al. 2020). This data was used as input features for model training in a previous work and, therefore, was adequately prepared for usage. This first part of the input data was enriched by collected data using the F1DataFetcher and PostgreSQL. Specifically, we collected data from the 2020 and 2021 seasons using our F1DataFetcher and the FastF1 API. This period was chosen due to the regulations in Formula 1 for these seasons. Specifically, the 1.6-litre V6 hybrid turbocharged engine type (FIA 2011) was used for each car. The new data contained granular information on 39 different Formula 1 Grand Prix and required restructuring to match the shape and metrics of the existing database. After a whole set of preparation steps, both data frames matched their structure. We then combined them into one.

5.2.2. Race Simulation

Since we needed to evaluate our trained model somehow, we took the race simulation provided by (Heilmeier, Graf, and Lienkamp 2018). This race simulation, which is publicly accessible via Github, is quite advanced and can, at least to some extent, simulate the real world. The race simulation works for several races from the seasons 2014 to 2019. External factors and strategic decisions are defined for each race individually (Heilmeier, Graf, and Lienkamp 2018). The simulation calculates lap times for each driver for each lap, sums them per driver and then returns the position and strategic decisions for each driver. Effects such as mass reduction due to fuel burning and tire degradation are also included in the model, which makes the whole set-up quite detailed and complex. Besides that, probabilistic influences are included and can be applied to the Monte Carlo simulation for each race (Heilmeier, Thomaser, et al. 2020).

5.2.3. Data Filtering

The combined data needed cleaning and filtering to be usable for the training of machine learning models. A group of facts and assumptions served to implement a filtering process for the raw data:

- **Exclusion of Grand Prix with missing lap times:** Some races collected needed more information on lap times. Some techniques were evaluated by trying to fill empty values by calculating the lap time from other timestamp data or using existing functions to impute missing values. The nature of motorsport, where seconds or fractions of a second are at stake, meant that these methods were not applicable. Therefore, races with missing timing data were eliminated from the raw data.
- **Exclusion of data for races in which it has rained:** Like the work of Heilmeier, Thomaser, et al. (2020), this work only examines data for Formula 1 races in stable, dry weather conditions.

Data for any race in which it has rained was removed from the data frame. Besides that, data for laps driven on intermediate or wet compound types were also removed.

- **Exclusion of bad tactical strategies**: As mentioned by Heilmeier, Thomaser, et al. (2020), some strategies can be seen as not successful. To later prevent the model from making senseless decisions, some strategic decisions needed to be removed from the data. Based on previous works, the threshold was set to be position 15. Any strategy leading to a driver finishing a race in a position worse than 15 was considered unsuccessful. Any data on that was omitted.
- **Exclusion of no-pit-stop-laps:** Since this work aims to train a machine learning model to be capable of predicting compound decisions, only data for actual compound decisions is needed. A change of compound type typically happens only once or twice per race for each driver. Therefore, the amount of data shrank drastically when filtering was applied, and all data was removed, where no pit stop with a tire change happened.

After filtering the combined data set, a final data frame consisting of 3862 pit stops from 7 different Formula 1 seasons was obtained. We then used this data frame for all further steps, including preprocessing, feature engineering and model training.

5.2.4. Output Variable

The target variable of the proposed model is the compound type. There are three distinct possible values: SOFT, MEDIUM, and HARD. These categories denote the dry compound types from which a driver can choose. Pirelli provides the exact compound types for each race, differing by race and season. The categories do not reflect the exact compound type but the overall category of the compound type. There are five compound types in Formula 1 these days: the three mentioned above plus INTERMEDIATE and WET. Since the last two are usually only run in the rain, and rain races have already been removed from the data set, they do not play a role in this model. Another exception is the seasons 2014 and 2015, where the compound type HARD did not exist. The HARD compound type was introduced in the 2016 season. To summarize, the model performs a classification with three possible classes: SOFT, MEDIUM, and HARD - except for seasons 2014 and 2015, where the model can only choose between two compound types.

5.2.5. Feature Engineering

Feature engineering is a crucial step when developing machine learning models. New features can provide more relevant, informative, and interpretable inputs for the algorithms, improving their decision quality and reliability. Besides the existing features of the data frame, new features were developed and constructed. They were then part of a feature selection process to examine their importance and whether they should have been part of the input features for the model training. Before proceeding with the steps of the feature selection, a summary of each one of the newly created features and the reasoning behind it is presented in the following:

- *Difference in lap time*: The difference in lap time in seconds to the previous lap could indicate a decline in performance, which could be related to tire degradation and, therefore, call for a tire change.

- *Lap time to average*: The lap time of each driver is divided by the average lap time of all other drivers in this lap. This new metric should depict whether the driver is slower or faster than the average driver in the field. A worn-out tire could cause a slower lap time than average, and the compound might need to change.

- *Tire age to average*: The tire age of each driver's currently worn tire is divided by the average tire age of all drivers in the same lap wearing the same compound type. A tire age above the average tirage could mean that a driver is wearing his tire too long, and a change might be needed.

- *Change in Interval*: Interval is a value in the data frame that deflects the time in seconds a driver is behind the driver in front of him. The change in the interval is calculated as the difference between the interval and the interval from the previous lap. A high value could indicate a need for a tire change as the driver falls behind his direct opponent.

- *Change in Gap*: Similar to the interval, the gap is an existing value that captures the time in seconds a driver is behind the leading driver on position 1. The change in the gap is also similarly calculated as the difference between the current gap and the gap in the previous lap. A high value could indicate a need for a tire change as the driver falls behind the entire field.

- *Laps since the last pitstop*: A simple count for the laps between 2 pitstops. A high value could indicate a need for a pit stop and a tire change.

These five features were calculated and added to the data frame. The entire data frame was then taken for the process of feature selection.

5.2.6. Feature Selection

Feature selection is a common approach and essential to tuning a machine-learning model. The general idea is to improve the model significantly by identifying the essential features of a machine learning model in the decision-making process while eliminating the input features that do not help or distort the decision. This work is based on Heilmeier et al. (2020), who stated that they had already performed feature selection for their model and defined six input features. We decided to keep these features, labelling them as fixed. Besides these features already identified as the most critical input features by Heilmeier et al. (2020), the focus was on the newly created features. Different methods were examined to derive the most essential features from the new set of potential input features. To do so, we performed the following steps:

- **Statistical testing of potential input features**: In the first step, ANOVA was performed for numerical input features, while Chi^2-Test was performed for categorical predictors, to test statistical significance.
- **Feature Importance for Random Forest Classifier**: In the next step, a Random Forest Classifier was used to find the importance of each feature on the decision variable.
- **Multicollinearity check**: The final step was to check for correlation between the potential input features. The Variance Inflation Factor (VIF) was calculated for each single feature.
- **Correlation pairs**: In the final step, highly correlated feature pairs were searched for. The ones in a pair with less importance were removed from the data.

From these steps, the final input data frame was derived. Besides the "fixed" input features from Heilmeier et al. (2020), some newly created features were identified as input features for the model training process, while others provided no additional information. Once the feature selection steps were performed, we obtained the final input data frame. The initial 6 metrics that were seen as fixed input features were enhanced by 4 newly created metrics that were identified as helpful metrics for the model training process. A quick overview of the final data frame can be found in Table 10.

Table 7: Final Input Features for Compound Decision

Category	**Feature**	**Name**	**Data Type**
Existing Features	Race progress	race_progress	numerical
	Remaining tire changes	remainingtirechanges_nl	numerical
	Number of available dry compounds	no_avail_dry_compounds	numerical
	Location	track	categorical
	Used 2 compounds	used_2compounds_nl	boolean
	Compound	compound	categorical
New Features	Laptime difference	laptime_diff	numerical
	Tireage to average	tireage_relative_to_avg	numerical
	Interval change	interval_change	numerical
	Gap change	gap_change	numerical

5.2.7. Preprocessing

The next step on the way to model training was the data preprocessing. Most machine learning algorithms require numerical data only. In real-world scenarios such as Formula 1, this is only sometimes granted. Besides that, most models are sensitive to unscaled data and require some normalization or scaling before the model training process. A preprocessing pipeline was created for the data to pass through to tackle these issues.

In the first step of preprocessing, we divided the data into categorical and numerical features. Categorical features were then transformed using scikit-learn's (Pedregosa et al. 2011) *OneHotEncoder*. This step creates binary columns for each category in a categorical column, filling the values with 0 or 1, where 0 denotes false, and 1 represents true. Numerical columns were scaled using scikit-learn's (Pedregosa et al. 2011) StandardScaler. The StandardScaler applies a z-score scaling to the data. This scaling mechanism converts data into a scale with a standard deviation of 1 and a mean of 0. Besides both transformation steps, scikit-learn's *SimpleImputer* was added to the preprocessing pipeline before transformation to fill empty data. The *SimpleImputer* was set to fill in numerical data with the mean and categorical data with the most common category.

5.2.8. Model Selection

In the model training phase, four machine learning models that predicted compound decisions were trained and evaluated against each other. As Heilmeier, et al. (2020) present in their work, a feed-forward neural network for the compound decision already exists. Therefore, we chose a different approach. The four machine learning algorithms chosen were Logistic Regression, Support Vector Machine, Random Forest Classifier and XG Boost. Each one of the models is briefly explained in the following:

- **Logistic Regression:** Logistic regression is a versatile and widely used statistical method. It is mainly used for binary outcome prediction but can be adapted to predict more than two categorical classes (Karsmakers, Pelckmans, and Suykens 2007).
- **Support Vector Machine:** Support Vector Machines are supervised machine learning algorithms that are primarily used for classification tasks. They operate by finding a hyperplane that best divides a set of data points into different classes. These models are commonly known for their capability of handling various data types and their effectiveness in high-dimensional data (Zhang 2012).
- **Random Forest Classifier:** A Random Forest Classifier is a supervised machine learning algorithm that applies multiple decision trees and combines their results (Kulkarni, Sinha, and Petare 2013). It is commonly known that this approach can significantly improve accuracy and perform very well on unseen data.
- **XG Boost:** XG Boost stands for extreme gradient boosting and is an efficient and scalable implementation of the boosting framework. As an advanced ensemble learning technique, XG Boost is commonly known for its speed and accuracy in machine learning problems (Kulkarni, Sinha, and Petare 2013).

All models were built and trained with the preprocessed input features. In the next step, we evaluated the different models.

5.2.9. Evaluation Metric

Several metrics aim to capture and represent model performance. The most common are accuracy, precision, recall, and F1-score. Each has its advantages and disadvantages, depending on the use case. We chose accuracy as our evaluation metric for our model, which has balanced input data. Accuracy measures how many predictions are correct. It is the proportion of accurate results among all examined predictions.

$$Accuracy = \frac{Number\ of\ correct\ predictions}{Total\ number\ of\ predictions} \tag{2}$$

This metric is also proposed by Heilmeier et al. (2020). Using the same metric as the previous work could also be beneficial as a comparison to the previously developed model can be performed. Besides accuracy, no other evaluation metric value is known from the neural network trained by Heilmeier et al. (2020).

5.2.10. Hyper Parameter Search

The different types of models presented above were trained, tested, and evaluated in an iterating process to define the optimal set of hyperparameters for each of them. An automatic approach of hyperparameter tuning was chosen by applying scikit-learn's (Pedregosa et al. 2011) *RandomizedSearchCV.* This process involves searching through specified subsets of hyperparameters for any given model and data set. It also uses k-fold cross-validation along the searching process, where data is automatically split into training and validation subsets 'k' times. Here, we set 'k' to 10, resulting in a 10-fold cross-validation, meaning the models were trained and evaluated on ten different subsets of data. The number of parameter settings was set to 100, meaning 100 different possible parameter combinations. This leads to 10 training phases for 100 different models of the same type, totalling 1000 training procedures. The *accuracy* of the performance was calculated for the model with the optimal hyperparameter settings. The results of the hyperparameter search and training for each model introduced earlier are presented in Table 11.

Table 8: Results of Hyperparameter Search

Model	Parameter	Accuracy
Logistic Regression	Solver: liblinear	0.6549
	C: 4.2813	
	C: 6.1460	
	Gamma: scale	
	Kernel: rbf	
Support Vector Machine	C: 6.1460	0.7597
	Bootstrap: False	
	Max Depth: 18	
	Min Samples Leaf: 1	
	Min Samples Split: 4	
Random Forest	N Estimators: 22	0.7813
	Max Depth: 7	
	Colsample bytree: 0.6531	
	Learning Rate: 0.0768	
	Subsample: 0.9635	
XG Boost	N Estimators: 197	0.7658
	Min child weight: 1	

Comparing these results to the feed forward neural network introduced by Heilmeier et al. (2020) which achieved an accuracy of 0.77 for the best fit, we can derive that our random forest model reached a slightly higher accuracy. Therefore, the Random Forest Classifier was implemented in the race simulation.

5.2.11. Model Deployment

To evaluate our model in an environment that is some-what close to the real world, we used the race simulation developed by Heilmeier et al. (2018) and explained above. We adjusted the feature calculation in the logic of the race simulation so that it would incorporate our newly created features that our trained model depends on. We replaced the preprocessing step in the simulation with our own preprocessing pipeline. Finally, we removed the neural network for compound decision (Heilmeier, Thomaser, et al. 2020) and included our own *Random Forest Classifier.* After these steps were conducted, we were set with our own race simulation to collect results and compare them to real world scenarios.

5.3. Results

Once the race simulation was set up and changed to our own requirements, we needed to simulate races under different circumstances and compare the results. We first thought about different scenarios which will be presented and explained in the following. We then defined a subset of drivers and races that we wanted to investigate further which will be explained later.

5.3.1. Scenarios

To be capable of performing a meaningful analysis of our simulation results, we thought about different scenarios for the drivers. The race simulation offers the possibility to apply the real strategies to the simulation or to use the newly created model for the compound decision. Both things can be applied individually for each driver. Based on this we created the following scenarios:

- *Scenario 1* (or *Base Scenario*): The race is simulated with each driver using the real strategy.
- *Scenarios 2-5:* In each scenario, one specific driver is using the introduced model.
- *Scenario 6:* In this scenario, each driver is using the introduced model.

Each of these scenarios is conducted for a subset of races. The results taken from these simulation runs are presented and compared in the following.

5.3.2. Simulation Procedure

Before starting the simulation for the scenarios defined earlier, we selected four drivers that represent the cluster they are in from the predefined cluster analysis. These four drivers were Kevin Magnussen (MAG), representing drivers in cluster 0; Lewis Hamilton (HAM), representing cluster 1; Daniel Ricciardo, as a reference to cluster 2; and George Russell (RUS), for cluster 3.

Next, we selected a few races of the 2019 season where all four drivers remained on the track. These races were: the Hungarian Grand Prix in Budapest, the Canadian Grand Prix in Montreal, the Japanese Grand Prix in Suzuka, the Austrian Grand Prix in Spielberg, the Abu Dhabi Grand Prix in Yas Marina Island, and the United States Grand Prix in Austin. These races were simulated with the different scenarios being applied to the drivers. We performed the simulation, including probabilistic influences per race. Each simulation was performed 5000 times using a Monte Carlo simulation, which is a functionality of the simulation (Heilmeier, Thomaser, et al. 2020). However, once the Monte Carlo simulation is applied, the simulation returns an average of the positions achieved in each race for each driver. There needs to be more information on the detailed strategic decisions. To retrace those decisions, one can simulate one race at a time without Monte Carlo simulation. We will mainly focus on the positional changes using Monte Carlo simulation to achieve more stable and reliable results. However, we will have a closer look at strategic decisions in some of the races. The results of the Monte Carlo simulations were compared to each other with one exception: The results of the base scenario are compared to the actual results of the race. All the other scenarios are compared to the simulation where the actual strategies are applied.

5.3.3. Simulation Results

We conducted the simulation for all the scenarios mentioned. The first one was to evaluate the simulation with actual strategies applied. Here, some differences in final positioning can occur in comparison to the real world due to the nature of the simulation. We have calculated the average deviation in positioning across the different races, shown in Figure 6.

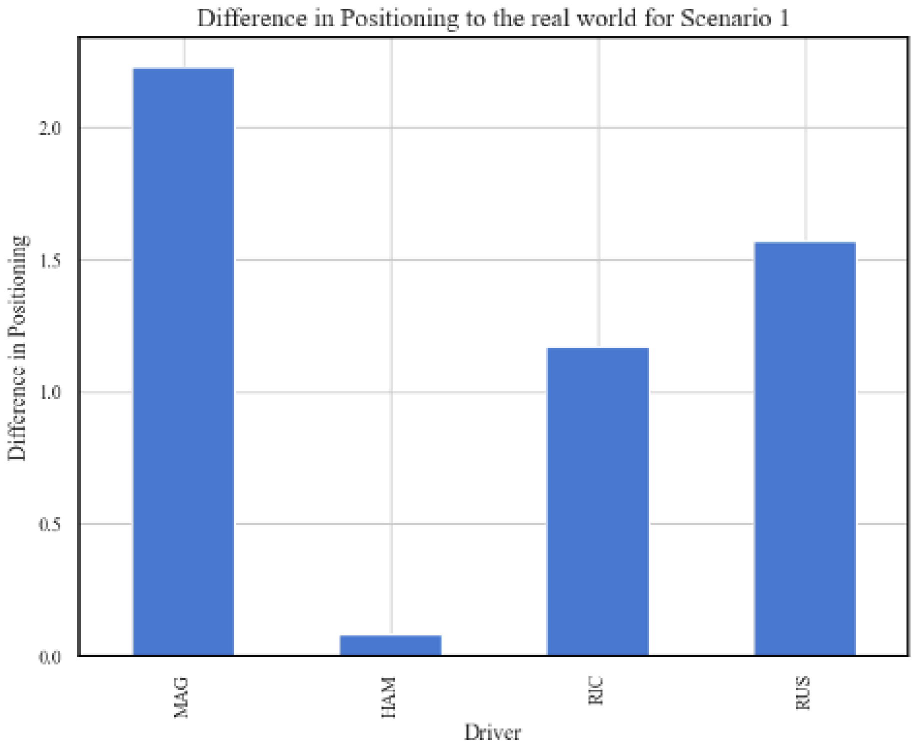

Figure 4: Average Positioning Difference for Scenario 1

As the plot shows, there are quite some differences in how close the simulation is to the actual results. While the simulation does not vary too much from the actual results for drivers like Hamilton (HAM), the difference for drivers like Ricciardo (RIC) and Russel (RUS) is more than one position per race. For Magnussen (MAG), the average difference exceeds two positions per race. These differences must be stated before we compare our model results to the simulation with scenario 1. Next, we simulated the races for scenarios 2 to 6.

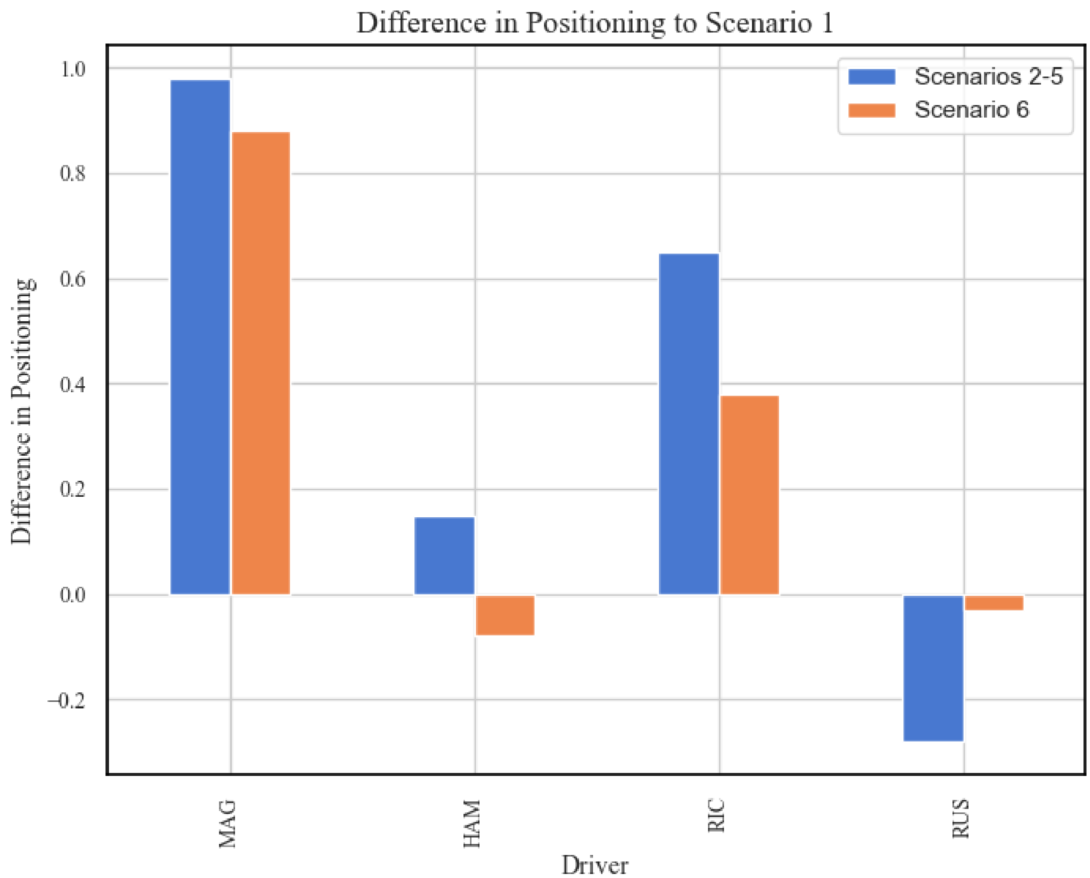

Figure 5: Average Positioning Differences for Scenarios 2-6

Figure 7 shows that the model worked quite well for 2 of the selected drivers, while it had little impact on the other 2 drivers. Deriving from that, cluster 0 and cluster 2, represented by Magnussen (MAG) and Ricciardo (RIC), seem to improve when applying the newly created model to their compound decision strategy. Drivers from clusters 1 and 3, represented here by Hamilton (HAM) and Russell (RUS), seem to not be impacted much by a change in compound decision suggested by the model. A detailed view how the model's strategies vary from the one made in the real race are evaluated for the Hungarian Grand Prix in Budapest and the United States Grand Prix in Austin. Compared to the actual strategy, the model suggested different compound strategies for all drivers in each race except for Russel (RUS) in the Hungarian Grand Prix. Interestingly, the model suggests different numbers of compound choices for Hamilton (HAM) in both races. Considering these strategic decisions, we can try to analyze and understand the model's impact on strategic decisions for each specific driver type.

Table 9: Detailed Compound Decision Comparison

Grand Prix	Driver	Real Strategy	Suggested Strategy
Austin	MAG	MEDIUM - HARD - SOFT	MEDIUM - MEDIUM - SOFT
	HAM	MEDIUM - HARD	MEDIUM - MEDIUM - HARD
	RIC	SOFT - HARD	SOFT - MEDIUM
	RUS	MEDIUM - HARD - SOFT	MEDIUM - HARD
Budapest	MAG	MEDIUM - SOFT	MEDIUM - HARD
	HAM	MEDIUM - HARD - MEDIUM	MEDIUM - HARD
	RIC	HARD - SOFT	HARD - MEDIUM
	RUS	MEDIUM - HARD	MEDIUM - HARD

5.3.4. Interpretation of Results

Taking the outcome of the clustering process of Formula 1 drivers into account, we can observe some meaningful insights. In this case, drivers from cluster 1, represented by Hamilton (HAM), are already at the top of their game and perform exceptionally well in each race. The model did not offer substantial improvements in race strategy. This makes sense; they are already excellent at managing race pace and pit strategies. Overall, there is little room for improvement. Drivers that belong to cluster 3, represented by Russell (RUS), are the drivers that are more cautious and generally could perform better. Their behaviour on the track does not align closely with the model's suggestions. The model was only trained with data for pit strategies that resulted in a final position better than 16. The model's highest impact was tracked for drivers in cluster 0, represented by Magnussen (MAG). These drivers prioritize endurance and tactical positioning, focusing on durability and long-term race planning. The model's suggestions have offered more effective tire strategies that align well with the existing strengths of drivers in this cluster. Drivers in cluster 2, represented by Ricciardo (RIC), have similar strengths as drivers in cluster 0, with different pit-stop decisions. They tend to have fewer pit stops and prefer hard compound tires. These preferences leave less room for the model to make decisions compared to cluster 0. Still, the model aligns quite well with the decisions made by drivers in cluster 2, and an improvement for this cluster can also be detected. In summary, the simulation results offer valuable insights into the potential of targeted, cluster-specific modelling in Formula 1. This approach holds promise for enhancing strategic decision-making, particularly in optimizing compound strategies in alignment with driver profiles and race dynamics.

5.3.5. Outlook

One thing comes to mind immediately when looking at the results. Training distinct models for different driver clusters could significantly improve the model decisions and their impact on race outcomes. Drivers in different clusters might need different models to make successful automated decisions in compound strategy. Besides that, future data could be beneficial for training and future research.

6. Discussion

In the initial phase of our study, we explored the application of advanced analytics in Formula 1, focusing on how driver clustering might inform pit-stop strategies. Our methodical approach to clustering revealed four distinct driver categories within the dataset. This classification emerged from a comprehensive analysis incorporating three key dimensions: performance, tactical, and behavioural patterns. These dimensions encompassed a variety of metrics, each contributing to a deeper understanding of driver profiles.

Despite the high-performance level uniformly exhibited by the drivers, which led to some similarity in cluster characteristics, our analysis succeeded in distinguishing between nuanced aspects of driver behaviour and strategy. The resulting profiles offered four differentiated interpretations of driver profiles and decision-making styles.

Subsequently, these clusters were instrumental in underpinning various predictive models and statistical analyses related to pit stop strategies. Our investigation spanned several critical areas, including optimal pit timing, tire selection, driver prioritization, and the influence of safety car periods. Through this analysis, we established associations between the divergent driver profiles and the outcomes of our analytical models. These connections allowed us to attribute specific strategic decisions and outcomes to identifiable cluster characteristics, thereby providing a deeper insight into the interplay between driver behaviour and pit stop strategy.

The research findings underscore the central role of driver clustering in enhancing the efficacy of analytical models within Formula 1. Notably, the application of driver clusters in our pit stop prediction models provided a more detailed understanding, enabling us to trace and explain strategic decisions in relation to specific driver characteristics.

These results from our study imply the critical importance of tailoring analytical models to reflect the distinct characteristics of drivers and teams in Formula 1. This customization is vital for capturing essential features and characteristics that might otherwise be overlooked in a broader, more generalized approach. Our findings suggest that the success of strategic models in Formula 1 hinges on their ability to account for the unique attributes and behaviours of individual drivers and teams.

This study, while thorough, encounters several limitations that should be considered when interpreting the results. Primarily, the availability of telemetry data through the FastF1 API, limited to the 2019 season onwards, restricts the temporal scope of our analysis of driver clusters and its evolution. Although future seasons will incrementally enrich our dataset, the absence of pre-2019 data constrains our historical analysis.

Additionally, a significant unknown in our study is the detailed information on car characteristics, which likely influences driver performance and tactics and ultimately the resulting clusters. The unavailability of these specifics, typically kept confidential by teams, may limit our understanding of the technical factors impacting driver behaviour. We attempted to mitigate this through feature engineering and normalization techniques but acknowledge that this is an approximation.

Another deliberate exclusion for our initial cluster analysis was races affected by rain. This decision was based on the disproportionate representation of wet races in our dataset, which could potentially skew the results. While this aids in maintaining data consistency, it omits the distinct strategic dynamics and behaviours prevalent in wet conditions. Furthermore, the challenge of quantifying track difficulty, despite having data on corners and marshal lights since 2019, presented another limitation. Our approach to normalising this data was a method to address this issue, but it remains uncertain.

These limitations underscore the need for cautious interpretation of our findings, particularly in their application to strategic decision-making in Formula 1. The clustering methodology applied for the 2019 to 2021 seasons and the resulting application on our advanced analytical models demonstrates promise for future applicability, yet it's important to consider these constraints and assumptions. Despite these challenges, we believe our methodology is robust and adaptable for future seasons.

Considering the dynamic nature of motorsport in general and Formula 1, which is subject to constantly evolving strategies, technologies and rules, our work is not intended to be a final solution, but rather food for thought for future work and analysis. As the scope of the data grows over time, there is the potential to deepen and develop our methodology. This promises even deeper and more differentiated insights into the strategic complexity of Formula 1 and motorsports. One key area is the expansion of the dataset. With the FastF1 API continually updating, incorporating data from subsequent seasons would not only improve the robustness of our clustering model but also facilitate longitudinal analyses. Such studies could track the evolution of driver strategies and team tactics over time. If telemetry data for seasons before 2019 becomes accessible, it would offer valuable historical insights, enriching our understanding of the sport's strategic development in response to technological and regulatory shifts.

Rain-affected races, excluded from our current analysis, represent a distinct strategic element in Formula 1. Future research could delve into these scenarios, perhaps through specialized models or by incorporating new features into existing frameworks. This focus could unveil how teams and drivers adapt to the challenges posed by variable weather conditions.

Another promising opportunity is the integration of track characteristics into the analysis. Understanding the influence of different track designs and surface conditions on driver performance and pit stop strategies would add considerable depth to our model. Additionally, while data on specific car builds is largely confidential, future collaborations with Formula 1 teams or utilization of public data could shed light on the relationship between car technology, driver skills, and strategic choices. Also, the field of data analytics and machine learning is rapidly advancing, offering the potential for refining our clustering model with more sophisticated algorithms. These advancements could handle larger and more complex datasets, providing finer-grained insights into driver performance and pit strategies.

As Formula 1 continues to evolve, so does the opportunity for deeper analytical exploration. The advancements in data collection and analysis present fertile ground for enriching our understanding of this complex sport. By adapting to these changes and refining our methodologies, we can contribute not only to academic discourse but also to the practical strategic toolkit of teams and drivers. This research represents a step towards a more data-driven and detailed comprehension of Formula 1 racing, setting the stage for future scholarly and practical advancements in the field.

7. Conclusion

This thesis has delved into the realm of Formula 1 pit stop strategies, employing advanced analytics to understand how driver clustering informs strategic decisions. We discovered four distinct driver categories based on performance, tactical, and behavioural dimensions, providing a nuanced view of driver profiles within the sport. This classification served as a foundation for examining pit stop strategies, precisely compound decisions in pit stop scenarios. The research illustrates the impact of driver characteristics on pit stop strategies in Formula 1.

This study enhances our comprehension of how individual behaviours and decisions intertwine with team strategies by linking divergent driver profiles to strategic outcomes. Using driver clusters in predictive models has shed light on the complexities of strategic decision-making in this high-speed sport. This thesis contributes to the understanding of Formula 1 racing by offering a data-informed perspective on the strategic elements of the sport. It underscores the value of bespoke strategies that consider the distinct qualities of drivers and teams, highlighting the potential of analytics in refining racing tactics.

In summary, this thesis provides an insightful exploration of the strategic dimensions of Formula 1 pit stops. It offers a clearer picture of how data analytics can be applied to decode the intricacies of racing strategies, enriching our understanding of this dynamic and technologically advanced sport.

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